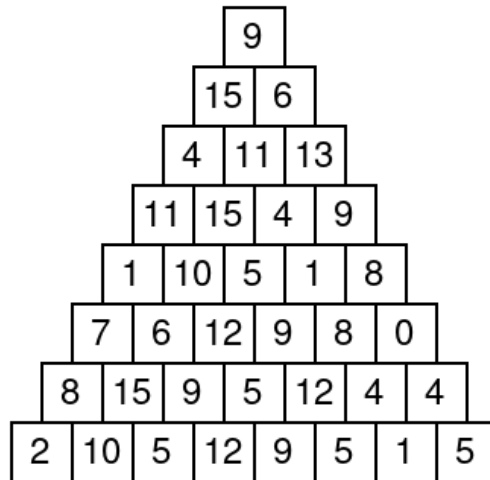


Xor Pyramid Diagonal

Time limit: 1.00 s

Memory limit: 512 MB

Consider a xor pyramid where each number is the xor of lower-left and lower-right numbers. Here is an example pyramid:



Given the bottom row of the pyramid, your task is to find the leftmost number of each row.

Input

The first line has an integer n : the size of the pyramid.

The next line has n integers $a_1, a_2, \dots, a_n$: the bottom row of the pyramid.

Output

Print n integers: the leftmost numbers of the rows from bottom to top.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a_i \leq 10^9$

Example

Input	Output
8 2 10 5 12 9 5 1 5	2 8 7 1 11 4 15 9